

```
In [1]: #Raw datato clean data conversion using Python EDA
```

```
In [2]: import pandas as pd
```

```
In [3]: pd.__version__
```

```
Out[3]: '2.2.3'
```

```
In [4]: #If any error then write, pip install --upgrade openpyxl
```

```
In [5]: emp = pd.read_excel(r'C:\Users\SID Balurkar\Desktop\Full Stack Data Science\Classwork\Nov 2025\21st Nov - EDA Workshc
emp
```

```
Out[5]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

```
In [6]: id(emp)
```

```
Out[6]: 2079748708336
```

```
In [7]: emp.columns
```

```
Out[7]: Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')
```

```
In [8]: emp.shape
```

```
Out[8]: (6, 6)
```

```
In [9]: emp.head()
```

```
Out[9]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year

```
In [10]: emp.tail()
```

```
Out[10]:
```

1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

```
In [11]: emp.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name         6 non-null      object
1   Domain        6 non-null      object
2   Age           4 non-null      object
3   Location      4 non-null      object
4   Salary        6 non-null      object
5   Exp           5 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes

```

In [12]: emp

Out[12]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [13]: emp.isnull()

```
Out[13]:
```

	Name	Domain	Age	Location	Salary	Exp
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	True	True	False	False
3	False	False	True	False	False	True
4	False	False	False	True	False	False
5	False	False	False	False	False	False

```
In [14]: emp.isna()
```

```
Out[14]:
```

	Name	Domain	Age	Location	Salary	Exp
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	True	True	False	False
3	False	False	True	False	False	True
4	False	False	False	True	False	False
5	False	False	False	False	False	False

```
In [15]: emp.isnull().sum()
```

```
Out[15]: Name      0
Domain    0
Age       2
Location  2
Salary    0
Exp       1
dtype: int64
```

```
In [16]: #Data Cleaning
```

```
In [17]: emp
```

```
Out[17]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

```
In [18]: emp['Name']
```

```
Out[18]: 0    Mike
1    Teddy^
2    Uma#r
3    Jane
4    Uttam*
5    Kim
Name: Name, dtype: object
```

```
In [19]: emp['Name'] = emp['Name'].str.replace(r'\W', '', regex = True)
```

```
In [20]: emp['Name']
```

```
Out[20]: 0    Mike
1    Teddy
2    Umar
3    Jane
4    Uttam
5    Kim
Name: Name, dtype: object
```

```
In [21]: emp.columns
```

```
Out[21]: Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')
```

```
In [22]: emp.head(1)
```

```
Out[22]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+

```
In [23]: emp['Domain'] = emp['Domain'].str.replace(r'\W', '', regex = True)
emp['Domain']
```

```
Out[23]: 0    Datascience
1         Testing
2    Dataanalyst
3        Analytics
4        Statistics
5           NLP
Name: Domain, dtype: object
```

```
In [24]: emp['Location'] = emp['Location'].str.replace(r'\W', '', regex = True)
emp['Location']
```

```
Out[24]: 0    Mumbai
1    Bangalore
2         NaN
3    Hyderbad
4         NaN
5        Delhi
Name: Location, dtype: object
```

```
In [25]: emp
```

Out[25]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34 years	Mumbai	5^00#0	2+
1	Teddy	Testing	45' yr	Bangalore	10%%000	<3
2	Umar	Dataanalyst	NaN	NaN	1\$5%000	4> yrs
3	Jane	Analytics	NaN	Hyderabad	2000^0	NaN
4	Uttam	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

```
In [26]: emp['Age'] = emp['Age'].str.replace(r'\W','',regex = True)
emp['Age']
```

Out[26]:

```
0    34years
1      45yr
2      NaN
3      NaN
4      67yr
5      55yr
Name: Age, dtype: object
```

```
In [27]: #The above age is still not correct. Use extract method.
```

```
In [28]: emp['Age'] = emp['Age'].str.extract('(\d+)')
emp['Age']
```

Out[28]:

```
0     34
1     45
2     NaN
3     NaN
4     67
5     55
Name: Age, dtype: object
```

```
In [29]: emp['Salary'] = emp['Salary'].str.replace(r'\W','',regex = True)
emp['Salary']
```

```
Out[29]: 0    5000
         1   10000
         2   15000
         3   20000
         4   30000
         5   60000
         Name: Salary, dtype: object
```

```
In [30]: emp['Exp'] = emp['Exp'].str.extract('(\d+)')
         emp['Exp']
```

```
Out[30]: 0     2
         1     3
         2     4
         3    NaN
         4     5
         5    10
         Name: Exp, dtype: object
```

```
In [31]: emp
```

```
Out[31]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [32]: clean_data = emp.copy()
```

```
In [33]: clean_data
```

```
Out[33]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [36]: #EDA technique - Remove missing values - 24th Nov -2025
```

```
In [37]: clean_data.isnull().sum()
```

```
Out[37]: Name      0
Domain    0
Age       2
Location  2
Salary    0
Exp       1
dtype: int64
```

```
In [39]: import numpy as np
```

```
In [48]: #Numerical data
```

```
In [40]: clean_data['Age'] = clean_data['Age'].fillna(np.mean(pd.to_numeric(clean_data['Age'])))
```

```
In [41]: clean_data['Age']
```

```
Out[41]: 0      34
          1      45
          2    50.25
          3    50.25
          4      67
          5      55
          Name: Age, dtype: object
```

```
In [42]: clean_data['Exp']
```

```
Out[42]: 0      2
          1      3
          2      4
          3    NaN
          4      5
          5     10
          Name: Exp, dtype: object
```

```
In [43]: clean_data['Exp'] = clean_data['Exp'].fillna(np.mean(pd.to_numeric(clean_data['Exp'])))
```

```
In [44]: clean_data['Exp']
```

```
Out[44]: 0      2
          1      3
          2      4
          3    4.8
          4      5
          5     10
          Name: Exp, dtype: object
```

```
In [45]: clean_data
```

```
Out[45]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50.25	NaN	15000	4
3	Jane	Analytics	50.25	Hyderbad	20000	4.8
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [ ]: #Categorical data
```

```
In [46]: clean_data['Location'] = clean_data['Location'].fillna(clean_data['Location'].mode()[0])
```

```
In [47]: clean_data['Location']
```

```
Out[47]: 0      Mumbai
1    Bangalore
2    Bangalore
3     Hyderbad
4    Bangalore
5         Delhi
Name: Location, dtype: object
```

```
In [49]: clean_data
```

```
Out[49]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50.25	Bangalore	15000	4
3	Jane	Analytics	50.25	Hyderabad	20000	4.8
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [50]: #Export this clean data from memory to system - csv
```

```
In [51]: clean_data.to_csv('clean_data.csv')
```

```
In [52]: import os  
os.getcwd()
```

```
Out[52]: 'C:\\Users\\SID Balurkar\\VSCODE_PROJECT'
```

```
In [53]: import matplotlib.pyplot as plt
```

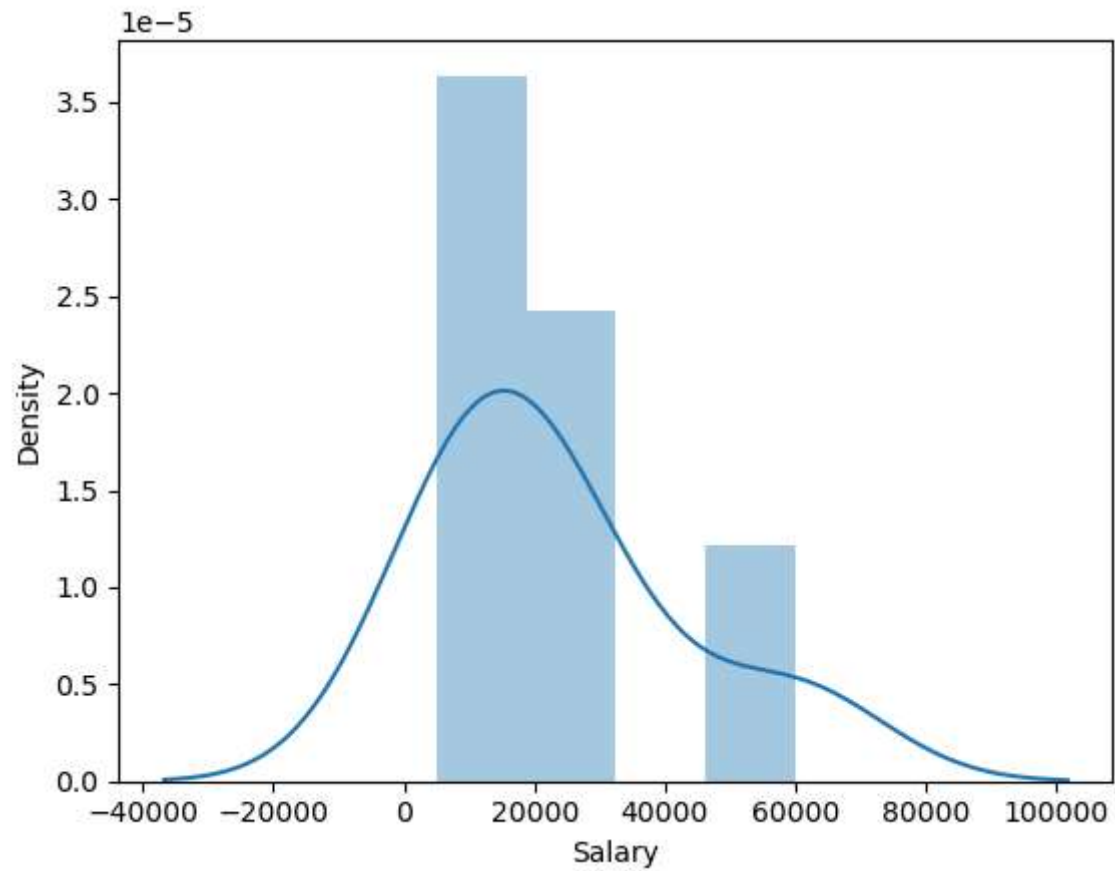
```
In [54]: import seaborn as sns
```

```
In [55]: import warnings  
warnings.filterwarnings('ignore')
```

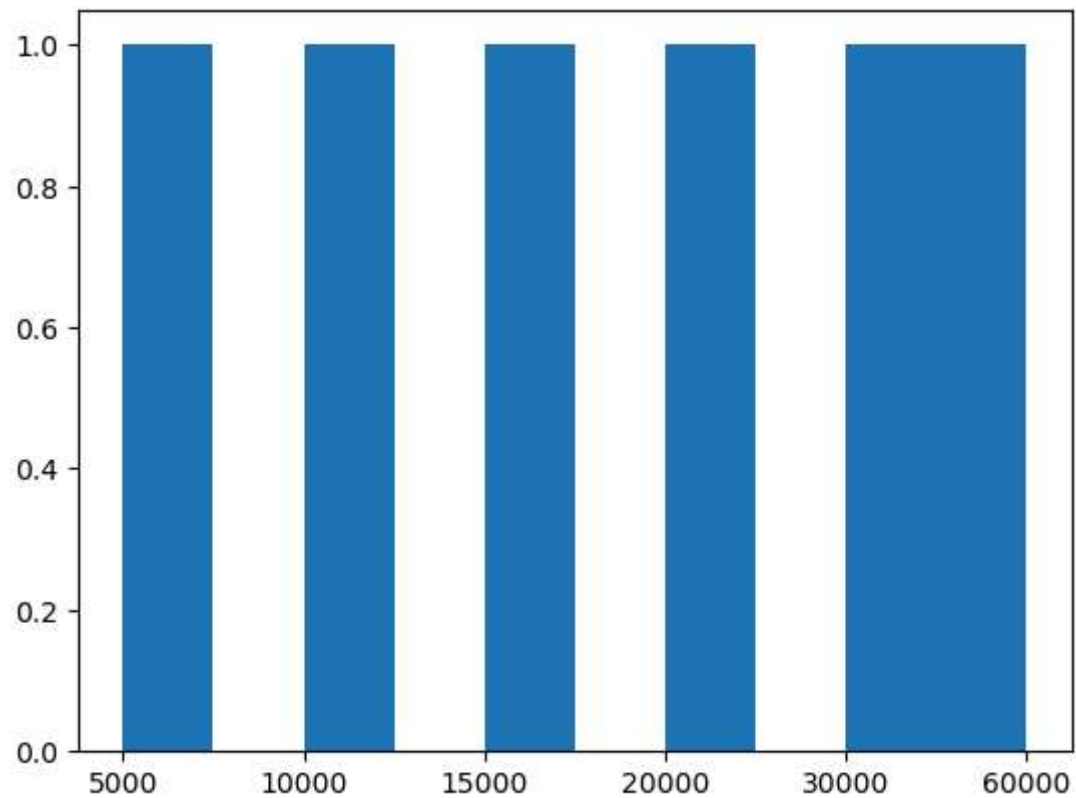
```
In [56]: clean_data['Salary']
```

```
Out[56]: 0      5000  
1     10000  
2     15000  
3     20000  
4     30000  
5     60000  
Name: Salary, dtype: object
```

```
In [57]: vis1 = sns.distplot(clean_data['Salary'])  
plt.show(vis1)
```



```
In [58]: vis2 = plt.hist(clean_data['Salary'])  
plt.show(vis2)
```



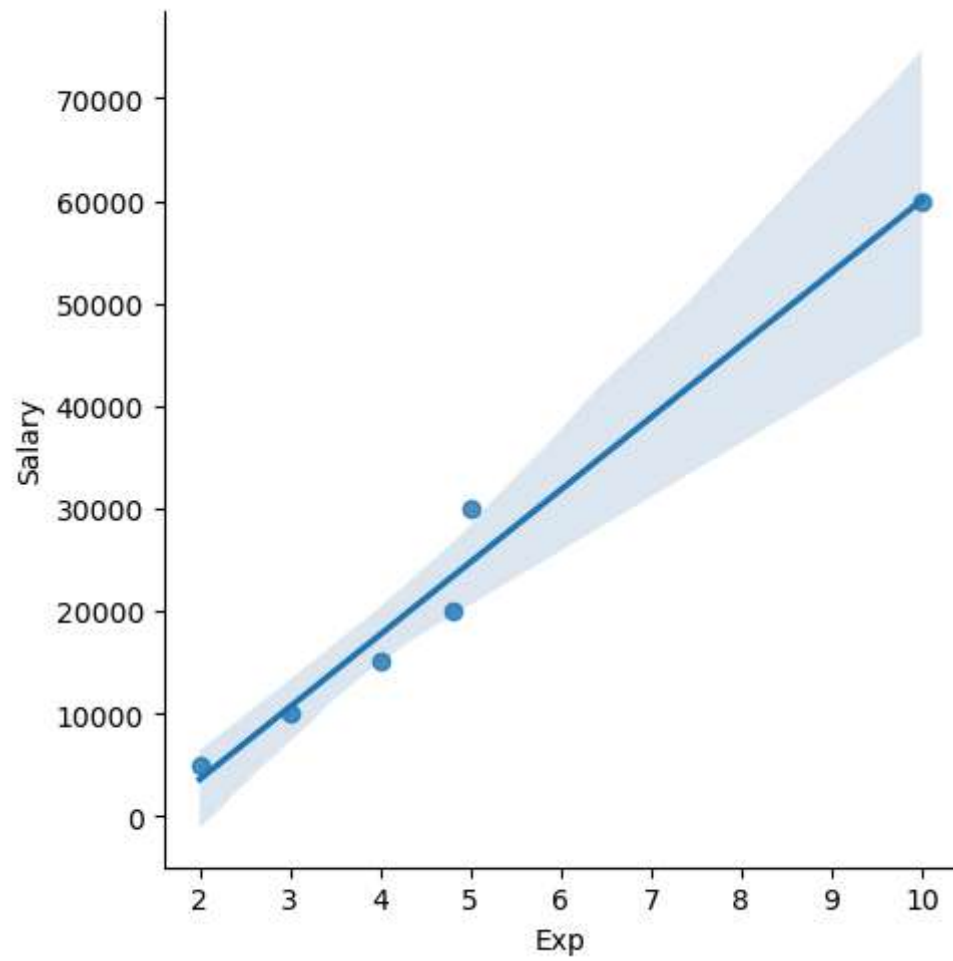
```
In [60]: #vis3 = sns.lmplot(clean_data x='Exp', y= 'Salary')
#plt.show(vis3)
```

Cell In[60], line 1

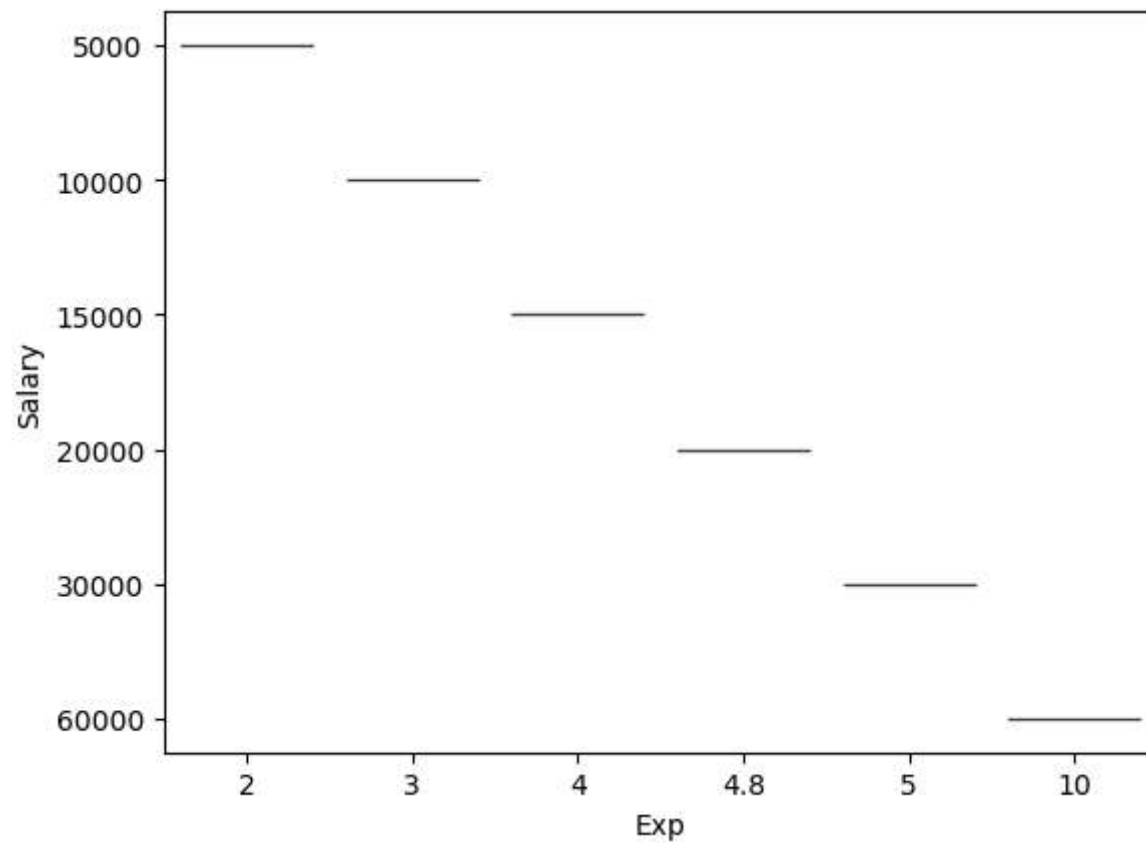
```
vis3 = sns.lmplot(clean_data x='Exp', y= 'Salary')
```

SyntaxError: invalid syntax. Perhaps you forgot a comma?

```
In [64]: clean_data['Salary'] = pd.to_numeric(clean_data['Salary'],errors = 'coerce')
clean_data['Exp'] = pd.to_numeric(clean_data['Exp'],errors = 'coerce')
vis3 = sns.lmplot(x='Exp', y= 'Salary', data = clean_data)
plt.show(vis3)
```



```
In [63]: vis4 = sns.boxenplot(data=clean_data, x='Exp', y='Salary')  
plt.show()
```



In [65]: `clean_data`

Out[65]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2.0
1	Teddy	Testing	45	Bangalore	10000	3.0
2	Umar	Dataanalyst	50.25	Bangalore	15000	4.0
3	Jane	Analytics	50.25	Hyderbad	20000	4.8
4	Uttam	Statistics	67	Bangalore	30000	5.0
5	Kim	NLP	55	Delhi	60000	10.0

```
In [66]: y_dv = clean_data['Salary']  
y_dv
```

```
Out[66]: 0      5000  
        1     10000  
        2     15000  
        3     20000  
        4     30000  
        5     60000  
        Name: Salary, dtype: int64
```

```
In [68]: clean_data.columns
```

```
Out[68]: Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')
```

```
In [67]: x_idv = clean_data[['Name', 'Domain', 'Age', 'Location', 'Exp']]  
x_idv
```

```
Out[67]:
```

	Name	Domain	Age	Location	Exp
0	Mike	Datascience	34	Mumbai	2.0
1	Teddy	Testing	45	Bangalore	3.0
2	Umar	Dataanalyst	50.25	Bangalore	4.0
3	Jane	Analytics	50.25	Hyderbad	4.8
4	Uttam	Statistics	67	Bangalore	5.0
5	Kim	NLP	55	Delhi	10.0

```
In [72]: clean_data
```

```
Out[72]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2.0
1	Teddy	Testing	45	Bangalore	10000	3.0
2	Umar	Dataanalyst	50.25	Bangalore	15000	4.0
3	Jane	Analytics	50.25	Hyderabad	20000	4.8
4	Uttam	Statistics	67	Bangalore	30000	5.0
5	Kim	NLP	55	Delhi	60000	10.0

```
In [69]: imputation = pd.get_dummies(clean_data, dtype = int)
imputation
```

```
Out[69]:
```

	Salary	Exp	Name_Jane	Name_Kim	Name_Mike	Name_Teddy	Name_Umar	Name_Uttam	Domain_Analytics	Domain_Datai
0	5000	2.0	0	0	1	0	0	0	0	
1	10000	3.0	0	0	0	1	0	0	0	
2	15000	4.0	0	0	0	0	1	0	0	
3	20000	4.8	1	0	0	0	0	0	1	
4	30000	5.0	0	0	0	0	0	1	0	
5	60000	10.0	0	1	0	0	0	0	0	

6 rows × 23 columns



```
In [70]: imputation.columns
```

```
Out[70]: Index(['Salary', 'Exp', 'Name_Jane', 'Name_Kim', 'Name_Mike', 'Name_Teddy',  
              'Name_Umar', 'Name_Uttam', 'Domain_Analytics', 'Domain_Dataanalyst',  
              'Domain_Datascience', 'Domain_NLP', 'Domain_Statistics',  
              'Domain_Testing', 'Age_50.25', 'Age_34', 'Age_45', 'Age_55', 'Age_67',  
              'Location_Bangalore', 'Location_Delhi', 'Location_Hyderabad',  
              'Location_Mumbai'],  
              dtype='object')
```

```
In [71]: len(imputation.columns)
```

```
Out[71]: 23
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```